King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Final Exam Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. $\int_1^4 \int_1^{e^2} \int_0^{\frac{1}{xz}} (\ln z)\, dy\, dz\, dx =$

(a) $\dfrac{\ln 4}{3}$

(b) $\ln 4$

(c) $3 \ln 4$

(d) $\dfrac{\ln 4}{2}$

(e) $2 \ln 4$

2. If $ax + by + cz = 7$ is an equation of the tangent plane to the ellipsoid $x^2 + 2y^2 + z^2 = 7$ at the point $(1, -1, 2)$, then $a + b + c =$

(a) 5

(b) -2

(c) -4

(d) 1

(e) 3

3. If $ax + by + cz = 10$ is an equation of the plane that passes through the points $(1, 2, 3)$, $(3, 2, 1)$, and $(-1, -2, 2)$, then $a + b + c =$

(a) 8

(b) -3

(c) 5

(d) 10

(e) -4

4. The volume of the parallelepiped having adjacent edges

$$\vec{u} = \langle 1, 3, 1 \rangle, \quad \vec{v} = \langle 0, 2, 2 \rangle, \quad \vec{w} = \langle 1, -2, 4 \rangle$$

is equal to

(a) 12

(b) 18

(c) 14

(d) 20

(e) 16

5. The minimum value of $f(x, y, z) = x^2 + 2y^2 + 3z^2$ subject to the constraint $x - 2y + 6z = 45$ is

(a) 125

(b) 105

(c) 95

(d) 135

(e) 115

6. The area of one petal (one loop) of the rose curve $r = 2\sin(5\theta)$ is equal to

(a) $\dfrac{2\pi}{5}$

(b) $\dfrac{3\pi}{10}$

(c) $\dfrac{3\pi}{5}$

(d) $\dfrac{\pi}{5}$

(e) $\dfrac{\pi}{10}$

7. $\int_{-4}^{4} \int_{0}^{\sqrt{16-x^2}} \cos\left(x^2 + y^2\right) dy\, dx =$

(a) $\int_{0}^{\pi} \int_{-4}^{4} r\cos(r^2)\, dr\, d\theta$

(b) $\int_{0}^{\pi} \int_{0}^{4} \cos(r^2)\, dr\, d\theta$

(c) $\int_{-\pi}^{\pi} \int_{0}^{4} r\cos(r^2)\, dr\, d\theta$

(d) $\int_{0}^{\pi} \int_{0}^{4} r\cos(r^2)\, dr\, d\theta$

(e) $\int_{0}^{2\pi} \int_{0}^{4} r\cos(r^2)\, dr\, d\theta$

8. The slope of the tangent line to the parametric curve

$$x = t^3 - 5t + 1, \quad y = t^2 + 3t$$

at the point corresponding to $t = 2$ is equal to

(a) $\frac{1}{3}$

(b) $\frac{7}{10}$

(c) $\frac{2}{9}$

(d) $\frac{5}{2}$

(e) 1

9. If (a, b) is the only critical point of $f(x, y) = x^2 + xy + \frac{1}{2}y^2 - 2x + y$, then

(a) f has a saddle point at (a, b)

(b) f has a relative maximum at (a, b)

(c) $f(a, b) = 5$

(d) $a + b = 1$

(e) f has a relative minimum at (a, b)

10. The graph of the surface represented by the cylindrical equation

$$r^2 \cos(2\theta) + z^2 + 1 = 0$$

is

(a) a sphere

(b) a hyperboloid of two sheets

(c) an elliptic hyperboloid

(d) a hyperboloid of one sheet

(e) an elliptic paraboloid

11. The function $f(x, y) = \dfrac{\sqrt{y - 2x}}{x^2 + y^2 - 4}$ is continuous on

(a) $\{(x, y) : x^2 + y^2 > 4,\ y \geq 2x\}$

(b) $\{(x, y) : -2 \leq x \leq 2,\ -2 \leq y \leq 2\}$

(c) $\{(x, y) : x^2 + y^2 \neq 4,\ y > 2x\}$

(d) $\{(x, y) : x^2 + y^2 \neq 4,\ y \geq 2x\}$

(e) $\{(x, y) : x^2 + y^2 \geq 4,\ y > 2x\}$

12. $\int_0^1 \int_0^y \left(6x + 5y^3\right) dx\, dy =$

(a) 0

(b) 1

(c) 4

(d) 2

(e) 3

13. The point with rectangular coordinates $(x, y, z) = (-2, 2\sqrt{3}, 4)$ is represented in spherical coordinates by

(a) $(\rho, \theta, \phi) = \left(4\sqrt{2}, \frac{5\pi}{3}, \frac{\pi}{4}\right)$

(b) $(\rho, \theta, \phi) = \left(2\sqrt{2}, \frac{2\pi}{3}, \frac{\pi}{4}\right)$

(c) $(\rho, \theta, \phi) = \left(4\sqrt{2}, \frac{2\pi}{3}, \frac{\pi}{4}\right)$

(d) $(\rho, \theta, \phi) = \left(4\sqrt{2}, \frac{2\pi}{3}, \frac{3\pi}{4}\right)$

(e) $(\rho, \theta, \phi) = \left(4\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4}\right)$

14. $\int_0^2 \int_{y^2}^4 f(x, y)\, dx\, dy =$

(a) $\int_0^2 \int_0^{\sqrt{x}} f(x, y)\, dy\, dx$

(b) $\int_0^2 \int_0^{x^2} f(x, y)\, dy\, dx$

(c) $\int_{y^2}^4 \int_0^2 f(x, y)\, dy\, dx$

(d) $\int_0^4 \int_0^{\sqrt{x}} f(x, y)\, dy\, dx$

(e) $\int_0^4 \int_1^{\sqrt{x}} f(x, y)\, dy\, dx$

15. Let $z = ye^x$. If (x, y) changes from $(2, 1)$ to $(2.1, 1.05)$, then using differentials the change in z is approximately equal to [Use the approximation $e^2 \approx 7.40$]

(a) 1.21

(b) 1.25

(c) 1.33

(d) 1.11

(e) 1.01

16. The arc length of the parametric curve

$$x = t^2, \quad y = \frac{1}{3}t^3 + 1, \quad 0 \le t \le 1$$

is equal to

(a) $\frac{2}{3}\left(5\sqrt{5} - 4\right)$

(b) $\frac{3}{4}\left(\sqrt{5} - 2\right)$

(c) $\frac{2}{3}\left(4\sqrt{5} - 8\right)$

(d) $\frac{1}{3}\left(5\sqrt{5} - 8\right)$

(e) $\frac{1}{3}\left(5\sqrt{5} + 6\right)$

17. $\int_0^2 \int_0^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} \sqrt{x^2+y^2+z^2}\, dz\, dy\, dx =$

(a) 4π

(b) π

(c) 8π

(d) 6π

(e) 2π

18. If $f(x,y) = \dfrac{x}{x+y}$, then $f_{yy}(x,y) - f_{xx}(x,y) =$

(a) $\dfrac{-2}{(x+y)^2}$

(b) $\dfrac{2(x-y)}{(x+y)^3}$

(c) $\dfrac{1}{(x+y)^3}$

(d) $\dfrac{2x}{(x+y)^3}$

(e) $\dfrac{2}{(x+y)^2}$

19. If $R = \{(x, y) : x^2 + y^2 \leq 9,\ x \geq 0,\ y \geq 0\}$, then $\iint_R (x + y)\, dA =$

(a) $\dfrac{2\pi}{3}$

(b) 6π

(c) 18

(d) 36

(e) 9

20. The average value of $f(x, y) = 2xy$ over the triangular region with vertices $(0, 0)$, $(1, 0)$, $(1, 3)$ is equal to

(a) $\dfrac{3}{2}$

(b) 2

(c) $\dfrac{5}{2}$

(d) $\dfrac{6}{5}$

(e) $\dfrac{1}{2}$

21. The volume of the solid lying in the first octant and bounded by the graphs of $z = 4 - y^2$, $y = x$, $y = 2$ is equal to

(a) 4

(b) $\dfrac{22}{3}$

(c) 6

(d) $\dfrac{32}{3}$

(e) $\dfrac{17}{3}$

22. The minimum distance from the point $(-1, -1, 0)$ to the surface $z = \sqrt{1 - 2x - 2y}$ is equal to (Hint: To simplify the computations, minimize the square of the distance).

(a) $\sqrt{2}$

(b) $\dfrac{\sqrt{3}}{2}$

(c) $\sqrt{3}$

(d) $2\sqrt{3}$

(e) $3\sqrt{2}$

23. The volume of the solid region bounded below by the paraboloid $z = x^2 + y^2$ and above by the sphere $x^2 + y^2 + z^2 = 6$ is given by

(a) $\int_0^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} \int_{x^2+y^2}^{\sqrt{6-x^2-y^2}} dz\, dy\, dx$

(b) $\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{\sqrt{6-x^2-y^2}}^{x^2+y^2} dz\, dy\, dx$

(c) $\int_0^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} \int_0^{\sqrt{6-x^2-y^2}} dz\, dy\, dx$

(d) $\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{x^2+y^2}^{\sqrt{6-x^2-y^2}} dz\, dy\, dx$

(e) $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{\sqrt{6-x^2-y^2}} dz\, dy\, dx$

24. The absolute maximum of $f(x, y) = x^2 - 4xy + 5$ on the region $R = \{(x, y) : 1 \leq x \leq 4,\ 0 \leq y \leq 2\}$ is

(a) -11

(b) 21

(c) 15

(d) -6

(e) 25

25. $\int_0^2 \int_x^2 x\sqrt{1 + y^3}\, dy\, dx =$

(a) $\frac{1}{3}\left(\sqrt{3} - 1\right)$

(b) 3

(c) $\frac{26}{9}$

(d) 9

(e) $\frac{26}{3}$

26. $\lim_{(x,y)\to(0,0)} \frac{2x^2-y^2}{x^2+2y^2} =$

(a) 0

(b) 2

(c) does not exist

(d) $-\frac{1}{2}$

(e) 1

27. Let $P(1,2)$ and $Q(2,3)$. The directional derivative of $g(x,y) = x^2 + y^2 + 1$ at P in the direction of the vector $\overrightarrow{PQ}$ is equal to

(a) $3\sqrt{2}$

(b) $4\sqrt{2}$

(c) 4

(d) 5

(e) $5\sqrt{2}$

28. Let $z = x^2 + y^2$, $x = s\cos t$, $y = \dfrac{\sin t}{s}$. The value of $\dfrac{\partial z}{\partial s}$ when $s = 2$ and $t = \dfrac{\pi}{4}$ is equal to

(a) $\dfrac{17}{4}$

(b) $\dfrac{5}{8}$

(c) $\dfrac{15}{4}$

(d) $\dfrac{3}{8}$

(e) $\dfrac{15}{8}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$\frac{\ln z}{xz} \to \frac{2}{x} \to 2\ln 4$.
2	**(d)**	Normal $\langle 1,-2,2\rangle$: plane $x-2y+2z=7$, so $a+b+c=1$.
3	**(c)**	Normal $\langle 4,-3,4\rangle$: plane $4x-3y+4z=10$, so $a+b+c=5$.
4	**(e)**	$V = \lvert\det\rvert = 1(12)-3(-2)+1(-2) = 16$.
5	**(d)**	$\lambda = 6$, point $(3,-3,6)$, $f = 135$ (the minimum).
6	**(d)**	One loop is $0 \le \theta \le \frac{\pi}{5}$: $A = \int_0^{\pi/5}(1-\cos 10\theta)\,d\theta = \frac{\pi}{5}$.
7	**(d)**	Upper half disk of radius 4: $\int_0^{\pi}\int_0^4 r\cos(r^2)\,dr\,d\theta$.
8	**(e)**	$\frac{dy}{dx} = \frac{2t+3}{3t^2-5}$, at $t=2$ equals $\frac{7}{7}=1$.
9	**(e)**	Critical point $(3,-4)$, $D=1>0$, $f_{xx}=2>0$: relative minimum.
10	**(b)**	$x^2-y^2+z^2=-1 \Rightarrow y^2-x^2-z^2=1$: hyperboloid of two sheets.
11	**(d)**	Need $y-2x\ge 0$ and $x^2+y^2\ne 4$.
12	**(d)**	Inner integral $3y^2+5y^4$, outer integral $=1+1=2$.
13	**(c)**	$\rho = 4\sqrt{2}$, $\theta = \frac{2\pi}{3}$ (second quadrant), $\phi = \frac{\pi}{4}$.
14	**(d)**	Region $0\le x\le 4$, $0\le y\le\sqrt{x}$.

Q	Ans	Brief explanation
15	**(d)**	$dz = e^2(0.1 + 0.05) = 0.15e^2 \approx 1.11$.
16	**(d)**	$L = \int_0^1 t\sqrt{4+t^2}\,dt = \frac{1}{3}(5\sqrt{5} - 8)$.
17	**(e)**	Octant of ball, $\rho \le 2$: $\int \rho^3 \sin\phi = 4 \cdot 1 \cdot \frac{\pi}{2} = 2\pi$.
18	**(e)**	$f_{xx} = \frac{-2y}{(x+y)^3}$, $f_{yy} = \frac{2x}{(x+y)^3}$, difference $\frac{2}{(x+y)^2}$.
19	**(c)**	Polar: $9\int_0^{\pi/2}(\cos\theta + \sin\theta)\,d\theta = 18$.
20	**(a)**	$\iint 2xy\,dA = \frac{9}{4}$, area $\frac{3}{2}$, so the average is $\frac{3}{2}$.
21	**(a)**	$V = \int_0^2 \int_x^2 (4 - y^2)\,dy\,dx = 4$.
22	**(c)**	$d^2 = x^2 + y^2 + 3$ is smallest at $(0,0)$: $d = \sqrt{3}$.
23	**(d)**	Disk $x^2 + y^2 \le 2$, $x^2 + y^2 \le z \le \sqrt{6 - x^2 - y^2}$.
24	**(b)**	No interior critical point; the largest boundary value is $f(4,0) = 21$.
25	**(c)**	Reverse order: $\int_0^2 \frac{1}{2}y^2\sqrt{1+y^3}\,dy = \frac{26}{9}$.
26	**(c)**	Along $y = 0$ the limit is 2, along $x = 0$ it is $-\frac{1}{2}$: does not exist.
27	**(a)**	$\nabla g(P) = \langle 2, 4\rangle$, $\mathbf{u} = \frac{1}{\sqrt{2}}\langle 1, 1\rangle$: $D_{\mathbf{u}}g = 3\sqrt{2}$.
28	**(e)**	$z_s = 2x\cos t - \frac{2y\sin t}{s^2} = 2 - \frac{1}{8} = \frac{15}{8}$.